**Assignment 3: Data Collection using TMDb API**

**d. Search for TV series**

Using the TMDb API, retrieve the **200 most popular TV series** released from 2015 onwards.

- Filter the results to include only TV series belonging to the **Drama** genre.
- Sort the results from most popular to least popular.
- Retrieve the following information:
    - TV series ID
    - TV series name
    - First air date
    - Popularity
    - Vote average
    - Vote count
- Save the results in tv_series.csv

**e. Find recommended TV series**

For each of the 200 TV series obtained in Question (d), use the TMDb API to retrieve the **5 recommended TV series**.

- If fewer than 5 results are returned, store all available results.
- Save the results in tv_series_similar.csv.

**f.** For each of the **300 movies obtained in Part (a)**, use the TMDb API to retrieve its corresponding **IMDb ID**.

Using the IMDb ratings dataset (title.ratings.tsv.gz), retrieve the following information for each movie:

- IMDb ID
- IMDb rating
- Number of IMDb votes
- IMDb URL

**g.** Combine the TMDb and IMDb information into a single DataFrame

**Compare TMDb and IMDb ratings**

Using the dataset created in Question (f):

1. Identify the **20 TV series with the highest IMDb ratings**.
2. Compare their IMDb rating with their TMDb vote average.

3. Calculate the difference:

$$\text{Rating Difference} = |\text{IMDb Rating} - \text{TMDb Vote Average}|$$

4. Display the TV series having the **largest difference** between the two ratings.
5. Create a scatter plot showing:

**X-axis:** TMDb vote average
**Y-axis:** IMDb rating

**h. Build a simple recommendation system**

Given the name of a TV series entered by the user:

1. Search for the TV series using the TMDb API.
2. Retrieve its TMDb ID.
3. Find its 5 similar/recommended TV series.
4. For each recommended series, retrieve its IMDb ID.
5. Display the recommendations in the following format:

Input Series: Stranger Things

Recommended Series

---

Series Name TMDb Rating IMDb Rating

---

Dark 8.4 8.7

The OA 7.5 7.8

...